

SOC10101 2026-7 TR1 003 Honours Project: Initial Project Overview (IPO)

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Title of Project:

Does Skill-Level Inference Improve Resume-to-Job Matching? An Evidence-Grounded Evaluation of Large Language Models.

Overview of Project Content and Milestones

Problem and motivation: Many organisations rank resumes with software and are adding large language models (LLMs) to this step. A resume rarely names every skill. A candidate who lists Python, scikit-learn and predictive modelling shows machine learning skill without writing the words, and keyword matching misses this person. Earlier work predicted such implicit skills with a convolutional network (91.34% precision and 98.79% recall on IT resumes) [1] and from similar job descriptions (mean reciprocal rank 29.4% above an explicit-skills baseline) [2]. LLMs offer a more direct route with new risks. GPT-4 ratings of 736 resumes correlated only minorly with human ratings [3], recent models still show bias linked to education [4], and LLMs invent skills. Whether LLM-inferred skills improve ranking or add noise is unclear.

Setting: TalentCLEF is a shared evaluation lab on skill and job title intelligence for human capital management. Its 2026 edition drew 113 registered teams and more than 400 submissions across Task A, ranking candidate resumes for a job, and Task B, retrieving ESCO skills for a job title [5]. The tasks run separately. This project predicts skills from the resume side and measures the effect on ranking. TalentCLEF 2027 adds an explainable ranking subtask, which checks whether claims are grounded in the source documents, and a task on ranking skills under negation [6]. Some 2026 participants built skill profiles from resumes for ranking, so their working notes need close reading [7].

Aim and stages: The project tests, in a controlled experiment, whether LLM-inferred skills help rank candidates for a job. Five stages cover the literature review, the experiment specification, the build of four matching setups in Python, testing with ranking metrics and statistics, and error analysis with the dissertation and poster.

Milestone	When
IPO agreed with Dr Md Zia Ullah, submitted on Moodle and emailed to Dr Gkatzia	By week 4
Draft report (aim, literature review, method outline) sent to Dr Dimitra Gkatzia	Before week 9
Interim meeting, then interim report submitted on Moodle	Week 9, 2 to 6 Nov 2026
Data and ethics settled, annotation guidelines written, prototype and two baselines built	End of December 2026
Remaining setups implemented, training and testing completed	End of January 2027
Evaluation and failure analysis completed	End of February 2027
Dissertation, poster and evidence submitted on Moodle	20 Apr 2027, 3pm

The Main Deliverable(s)

A dissertation reporting a controlled experiment on whether LLM-inferred skills improve candidate ranking. Supporting deliverables are a Python pipeline with four matching setups (code, prompts and results tables in a GitHub repository), a poster, and an interim report, which goes into the dissertation as an appendix.

The Target Audience for the Deliverable(s)

Dr Md Zia Ullah and Dr Dimitra Gkatzia assess the work. It also targets natural language processing and information retrieval researchers who study person-job fit and skill extraction, including the TalentCLEF community, developers of applicant tracking tools, and recruiters who need to know whether an LLM skill step is worth adding.

The Work to be Undertaken

Investigation compares prior work on skill extraction, implicit skills and LLM-based matching. Specification covers the data, the labels and the annotation guidelines. Implementation builds the setups below in Python. Evaluation runs them on held-out data. Analysis studies where the LLM goes wrong. The main question is whether evidence-supported LLM skill inference improves resume-to-job matching beyond document-level semantic matching and explicit skill matching.

- **RQ1:** How accurately does an LLM infer skills a resume supports but does not state?
- **RQ2:** Does adding inferred skills improve candidate ranking compared with both baselines?
- **RQ3:** How reliably does the quoted evidence support each inferred skill?
- **RQ4:** Which kinds of errors does the LLM make?

Four hypotheses guide the work, none assumed: evidence-supported inference finds relevant unstated skills (H1), improves ranking over explicit skills alone (H2), reduces unsupported skills compared with unconstrained inference (H3), and the gain differs by job family and skill type (H4). An inferred skill is one the resume supports through its content without stating it. Python alone does not support Kubernetes, and "worked on data projects" is too vague to support data science. Every inferred skill is labelled supported, unsupported or ambiguous against its quoted resume sentence. Comprehensive annotation is conducted internally, and a second annotator, agreed with Dr Md Zia Ullah, labels a subset so that Cohen's kappa is reported.

Setup	Representation	Explicit skills	Inferred skills	Evidence check
B1	Full-document semantic similarity	No	No	No
B2	Explicit skills only	Yes	No	No
B3	Explicit plus LLM-inferred skills	Yes	Yes	No
B4	Explicit plus inferred skills, validated	Yes	Yes	Yes

B3 against B4 is the main comparison. Skills on both sides map to ESCO [8] by embedding retrieval and LLM re-ranking, the design Clavié and Soulié used [9], with a hand check on the evaluation subset. Humans supply the relevance labels, never an LLM. TalentCLEF Task A comes first. Its data are synthetic, hold no personal information and come with relevance judgements [6]. Availability and terms of use need checking, and if the data do not fit, a smaller sample is labelled with a second annotator. Data are split into development, validation and test sets, with jobs from one family or template kept in one split, and prompts are locked before the test set.

Skill inference is scored with precision, recall and F1, and ranking with MAP, MRR, nDCG@10 and Recall@10. TalentCLEF uses MAP as its official metric and reports MRR [6]. Setups are compared with a paired Wilcoxon signed-rank test, a paired bootstrap 95% confidence interval and an effect size. Errors are sorted into groups such as unsupported inference, over-generalisation, synonym and hierarchy confusion, negation and soft-skill over-prediction. Negation gets its own test, in line with the 2027 negation task [6]. A smaller experiment changes names, gender cues and education details in a resume sample and measures how rankings move [4]. One open-weight model runs locally and is fixed after development. Every run records the model version, prompt, temperature, seed and date.

Additional Information / Knowledge Required

New knowledge is needed in text embeddings, structured prompting so the skill step returns a skill with its supporting quote, ranking evaluation and significance testing, annotation design and inter-annotator agreement, and fairness testing for hiring systems. Python, Jupyter Notebook and GitHub skills will be extended. The expected tools are a sentence embedding library, a locally run open-weight LLM, the ESCO files and a statistics library.

Information Sources that Provide a Context for the Project

Implicit skills: Jiechiew and Tsopze predicted high-level skills from IT resumes even when the resumes did not state them [1]. Gugnani and Misra took implicit skills from similar job descriptions and reached a mean reciprocal rank of 0.88 [2]. Goyal et al. built JobXMLC, which predicts skills missing from job descriptions and beats earlier methods by 6% in precision and 3% in recall [10]. Clavié and Soulié matched job text to ESCO with LLMs and no human annotation, and GPT-4 re-ranking improved RP@10 by more than 22 points [9].

LLM ranking and bias: Vaishampayan et al. found a minor correlation between zero-shot GPT-4 and human ratings of 736 resumes [3]. Iso et al. found that recent models reduced gender and race bias while bias linked to educational background stayed significant [4]. Yu et al. trained LLM re-rankers for person-job fit on noisy recruiting data and reported gains over GPT-5 and Claude Opus 4.5 [11].

Organisations: The European Commission runs ESCO [8], and the CLEF TalentCLEF lab publishes the shared tasks [5], [6]. Further papers will come from the ACL Anthology, arXiv, Google Scholar and IEEE Xplore. The intended users are recruiters who screen large applicant pools.

In the literature reviewed so far, no study has been found that evaluates LLM-inferred resume skills against both document-level semantic matching and explicit skill matching while requiring textual evidence for each skill. The search continues, and the claim will be adjusted if such a study appears.

The Importance of the Project

Keyword search misses candidates who describe their skills in other words, and teams are trying LLM screening on thin evidence. GPT-4 ratings differ from human ratings [3] and bias linked to education remains [4]. A clear answer on whether an inferred skill step adds value, and what it costs in errors, helps anyone deciding how to build or buy such tools. Implicit skill inference is not new [1], [2]. The novelty is the design. LLM inference is treated as an intervention in ranking, compared under control with both baselines, and every inferred skill is checked against evidence in the resume. The B3 against B4 ablation shows whether that check matters, and the design fits the grounded, explainable direction of TalentCLEF 2027 [6]. A local model keeps resume text off hosted services, and code, prompts and results tables will be shared so others reproduce the experiment.

The Key Challenge(s) to be Overcome

Ground truth is the hardest part. Public data rarely records who got hired, and real recruiting data is noisy [11]. Human relevance labels are needed from a shared task or a carefully built sample, and their limits will be stated. Synthetic TalentCLEF profiles differ from real resumes, which limits how far findings transfer. The model invents skills, so the evidence rule, hand labels and kappa check will take real time. Ethics approval and licence checks come before any real resume data is used. The project sits close to earlier implicit skills work [1], [2], so the report must say plainly what is new: the controlled comparison, the evidence check and the ablation. Sample sizes will be fixed early, and the project stays at four setups and one main model.

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